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DAA LAB

LAB EXPERIMENTS

EXPERIMENT – 1

OUT OF BOUNDARY PATHS

AIM:

To implement a Python program to determine the probability that a ball starting from a given cell in a grid will move out of the boundary after a specified number of moves.

ALGORITHM:

1. Initialize the grid dimensions, starting row and column, and the maximum number of moves.
2. Create a dynamic programming table to store the number of ways the ball can reach each cell.
3. For every move, check the four possible directions: up, down, left, and right.
4. If a move goes outside the grid boundary, increment the count of paths that leave the grid.
5. Otherwise, add the number of ways to the destination cell in the next table.
6. Repeat the process for all allowed moves.
7. Return the total number of paths that went outside the boundary.

PROCEDURE:

1. Open a Python environment such as Google Colab, Jupyter Notebook, or IDLE.
2. Define a function named out_of_boundary_paths().
3. Enter the number of rows, columns, maximum moves, and starting position.
4. Execute the program.
5. The program calculates the number of paths that leave the grid.
6. Display the final result.

PYTHON CODE:

```python
def find_paths(m, n, moves, r, c):
    count = 0
    for _ in range(moves):
        if r == 0:
            count += 1
        if r == m - 1:
            count += 1
        if c == 0:
            count += 1
        if c == n - 1:
            count += 1
        if r < m - 1:
            r += 1
        else:
            r -= 1
    return count
m = 2
n = 2
moves = 2
r = 0
c = 0
print("Number of paths:", find_paths(m, n, moves, r, c))
```

SAMPLE OUTPUT:

Number of paths : 4

```
Output

Number of paths: 4
```

RESULT:

Thus, the Python program to find the number of paths that move out of the boundary was successfully implemented and executed.